

Pokémon Red benchmark: Jev alone vs Jev + GPT-6 Sol

First comparison · 28 September 2026 · runs tagged v2

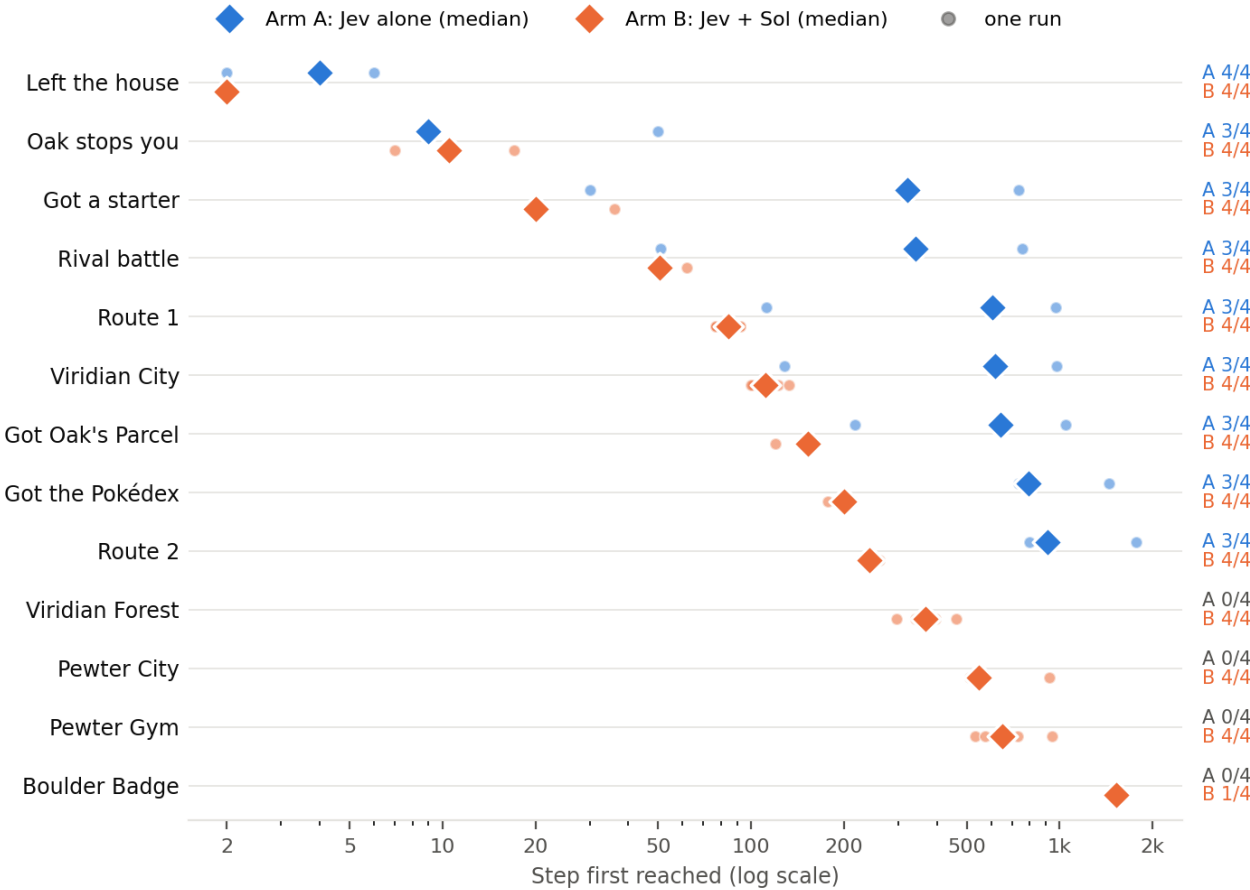
Both arms play from the same new-game save with the same harness. Jev (TypeSafe, via OpenRouter) makes every decision in both arms. In Arm A its goal is fixed (“make progress in the main story toward the next badge”); in Arm B, GPT-6 Sol reads the game state every 15 steps and rewrites that goal. Each run is capped at 2,000 steps and ends early after 1,000 steps without a new milestone.

	Arm A: Jev alone	Arm B: Jev + Sol
Average milestones (of 13)	7.0	12.2
Runs that beat Brock (badge 1)	0 of 4	1 of 4
Average blackouts per run	2.8	9.2
Average cost per run	\$0.09	\$0.62

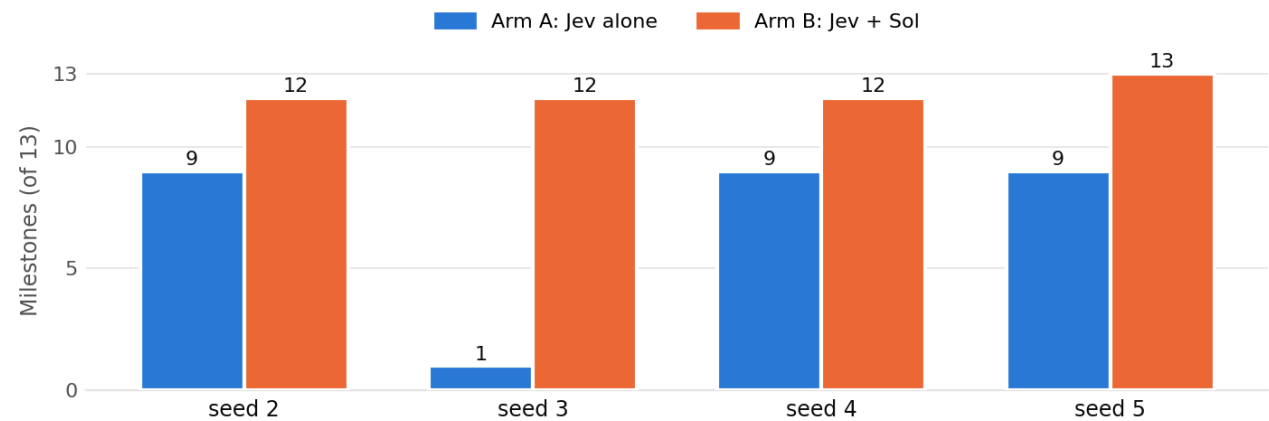
Seeds 2, 3, 4, 5 (4 runs per arm).

How fast each arm reached each milestone

Each small dot is one run; the diamond is the median. The column on the right counts how many runs reached that milestone at all. The step axis is logarithmic so the early milestones stay readable.



Milestones reached per run



Median step per milestone

Median over the runs that reached it; brackets show how many runs did. Speed-up = Arm A median ÷ Arm B median.

Milestone	Arm A median step	Arm B median step	Speed-up
Left the house	4 (4/4)	2 (4/4)	2.0×
Oak stops you	9 (3/4)	10 (4/4)	0.9×
Got a starter	320 (3/4)	20 (4/4)	16.0×
Rival battle	341 (3/4)	50 (4/4)	6.8×
Route 1	606 (3/4)	84 (4/4)	7.2×
Viridian City	616 (3/4)	112 (4/4)	5.5×
Got Oak's Parcel	642 (3/4)	152 (4/4)	4.2×
Got the Pokédex	791 (3/4)	200 (4/4)	3.9×
Route 2	912 (3/4)	242 (4/4)	3.8×
Viridian Forest	- (0/4)	368 (4/4)	-
Pewter City	- (0/4)	548 (4/4)	-
Pewter Gym	- (0/4)	652 (4/4)	-
Boulder Badge	- (0/4)	1527 (1/4)	-

Every run

Arm	Seed	Steps	Score	Starter	Rival	Blackouts	Ember uses	Cost
A	1*	2000	10	Charmander	lost	4	0	\$0.111
A	2	1798	9	Charmander	won	7	0	\$0.100
A	3	1006	1	-	-	0	0	\$0.056
A	4	2000	9	Squirtle	won	2	0	\$0.108
A	5	1912	9	Charmander	won	2	0	\$0.109
B	1*	2000	12	Charmander	won	1	3	\$0.771
B	2	1732	12	Charmander	won	7	15	\$0.592
B	3	1533	12	Charmander	lost	13	9	\$0.542
B	4	1947	12	Charmander	won	10	12	\$0.649
B	5	2000	13	Charmander	won	7	37	\$0.685

* Seed 1 ran before a map-name fix: its “Viridian Forest” milestone was really the forest gate, and Arm B’s “Pewter Gym” was really the museum’s second floor. It is excluded from the charts and averages above.